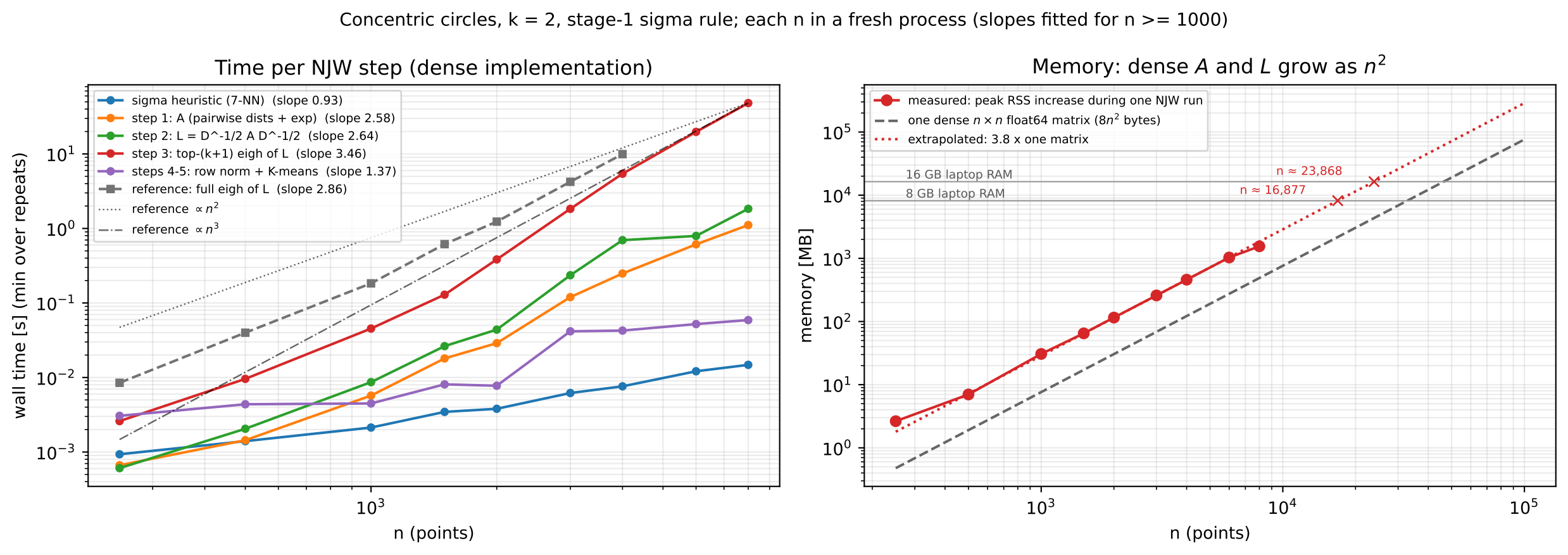
Concentric circles, k = 2, stage-1 sigma rule; each n in a fresh process (slopes fitted for n >= 1000)
Time per NJW step (dense implementation)
wall time [s] (min over repeats)
n (points)
sigma heuristic (7-NN) (slope 0.93)
step 1: A (pairwise dists + exp) (slope 2.58)
step 2: L = D^-1/2 A D^-1/2 (slope 2.64)
step 3: top-(k+1) eigh of L (slope 3.46)
steps 4-5: row norm + K-means (slope 1.37)
reference: full eigh of L (slope 2.86)
reference ∝ n²
reference ∝ n³
Memory: dense A and L grow as n²
memory [MB]
n (points)
measured: peak RSS increase during one NJW run
one dense n × n float64 matrix (8n² bytes)
extrapolated: 3.8 x one matrix
16 GB laptop RAM
8 GB laptop RAM
n ≈ 23,868
n ≈ 16,877